

Essential Mathematics and Statistics for Scientists

Carnegie Mellon University
School of Computer Science
Computational Biology Department

02-680 – Fall 2026

Revised: July 30, 2026
(change since start of classes marked in orange)

DRAFT

Class Location and Times:

[Lecture] Tuesday & Thursday 9:30am-10:50am,
Posner Hall (POS) 152

[Recitation] Friday 5:00pm-6:20pm,
Margret Morrison (MM) 103

Course Description: This course rigorously introduces fundamental topics in mathematics and statistics to first-year master's students as preparation for more advanced computational coursework. Topics are sampled from mathematical logic and set theory, graph theory, linear algebra (vectors, matrices, linear systems, vector spaces, eigendecomposition, and singular value decomposition), differential calculus and gradient-based optimization, probability distributions and densities, multivariate probability distributions and graphical/Markov models, maximum likelihood estimation, Bayesian (MAP) estimation, statistical inference, hypothesis testing, and linear regression. Students completing this course will obtain a broad skillset of mathematical techniques and statistical inference. They will have the quantitative foundation to immediately step into an introductory master's level machine learning or automation course. This background will also serve students well in advanced courses that apply concepts in machine learning to scientific datasets, such as 02-710 (Computational Genomics) or 02-750 (Automation of Biological Research). The course grade will be computed as the result of homework assignments, midterm tests, and class participation.

Prerequisite: High school mathematics (including single- and multi-variable calculus) and ability to reason mathematically.

Course Objectives: At the end of the semester, students will be able to:

1. Reason precisely using formal mathematical language as a foundation for describing computational and biological problems.
2. Perform and interpret core linear algebra operations as they apply to real datasets.
3. Use calculus, particularly derivatives and gradients, to analyze and optimize functions.
4. Apply the axioms of probability and reason about random variables, distributions, and multivariate/graphical models.
5. Estimate model parameters from data using maximum likelihood and Bayesian methods, and quantify uncertainty in those estimates.
6. Design and interpret statistical analyses, including hypothesis tests and regression models, and communicate their conclusions and limitations.

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1 Instructional Staff

Instructors

Dr. Dan DeBlasio email: deblasio@cmu.edu
 office: GHC 7707
 office hours: TBD
 appointments: calendly.deblasiolab.org

Teaching Assistants

2 Logistics

Course Topic Schedule: Can be found [on Google Sheets¹](#) and is subject to change.

Topics covered this semester:

- Logic
- Graph theory
- Phylogenetics*
- Set theory
- Sets and functions
- Linear algebra
- Matrices and vectors
- Probabilities
- Probability distributions
- Probability densities
- Multivariate distributions
- Maximum likelihood estimation
- Bayesian and Statistical inference
- Hypothesis testing

Textbook: There are no required textbooks for this course. Below are textbooks that can be good references:

- **Mathematics and Statistics for the Quantitative Sciences** by Betti
- **Connecting Discrete Mathematics and Computer Science** by Liben-Nowell²
- **Mathematics for Machine Learning** by Deisenroth, Faisal, Ong
- **All of Statistics: A Concise Course in Statistical Inference** by Larry Wasserman
- **Probability and Statistics** by Morris DeGroot and Mark Schervish
- **Basic Concepts of Linear Algebra** by Isaak and Manougian

Communication platforms:

- **Canvas Homepage.** The course homepage will be hosted on Canvas. Canvas will be used for attendance and as a central repository for grades. You should be automatically enrolled at <https://canvas.cmu.edu/courses/54762>.
- **Discussion Forum.** An online forum is provided on Piazza as an area for discussion and questions. The forum will be moderated by the course staff who will respond to questions, but students are encouraged to help each other via discussion. However, assignment specifics should not be discussed — any hints will be provided by the teaching staff. You can find the class on Piazza at <https://piazza.com/cmu/fall2026/02680>.

3 Grading

Grades are communicated to students in a timely manner. It is the students' responsibility to keep track of their grades by compiling the grades they receive. Your semester

¹<https://docs.google.com/spreadsheets/d/1AuxF0rYH08UFzA6YwOEVaYlRdU7XbZr-j7E7Gvi0G9s>

²<https://www.cs.carleton.edu/faculty/dln/book/>

grade will be based on a combination of homework assignments, weekly quizzes, class participation, mid-term assessment, and a final exam.

The approximate percentages are as follows:

50%	Homework
10%	Attendance & Participation
40%	Exams (20% per)

The base percentage-score-to-letter-grade conversion for this course is as follows:

	+		-
A	98%	93%	90%
B	88%	83%	80%
C	78%	73%	70%
D	68%	63%	—

Any lower scores will result in a grade of “R”. These minimums may be lowered without notice but will not be raised.

3.1 Homework

Written homework assignments will test your knowledge of the material covered in class. Homework assignment will be made available at least one week before the due date.

All writing assignments are to be submitted on Canvas (via Gradescope) on the day it is due. Any assignment turned in after the due date will be deducted 5 percentage points for the first day it is late, and 20 percentage points for every day after that. If you have extenuating circumstances, just communicate with us and we'll do what we can to accommodate.

Homework will be submitted as a pdf document which will need to be generated using an editor of the student's choosing. Due to the reliance of this course on mathematical notation the use of $\text{\LaTeX}$ is encouraged. If using another editor, students should utilize the editor's equation editor to ensure the correct answers are being communicated. To assist those students using $\text{\LaTeX}$, the source code for each assignment will be provided in addition to the pdf.

Cheating policy

All work must be your own. Unauthorized collaboration or plagiarism will result in a negative grade (e.g., a homework worth 100 points will be factored in as a -100 points towards your final grade) and will be reported to your academic advisor and dean.

See Section 5.

3.1.1 Regrading of Assignments

In the event that you feel an assignment was graded incorrectly, you must make the request in writing within one week of receiving the graded material. Your request should include a reference to the section of the assignment you believe was incorrectly graded. By asking for your assignment to be regraded, the whole assignment will be regraded and it is possible that fewer points may be awarded than the previous grade in light of new consideration.

3.2 Attendance & Participation

Attendance will *not* be taken, but we will have occasional in-class exercises that serve to reinforce the concepts we have covered and provide feedback to the instructor. These exercises will not be graded for correctness, but participation will be expected in order to receive credit. You are allowed three “dropped” attendance grades without penalty. These can be used for any purpose.

All classes will be live without an option for virtual participation.

3.3 Exams

The two exams will test your knowledge of the material from the class. The first midterm will be held in class, and the second midterm will be held during the university’s scheduled time. Dates for the exams will be announced in class, and can be found on the course schedule when available.

The exams will not be cumulative: midterm 2 will cover material encountered after midterm 1. That having been said, later material in the class may build upon the techniques covered in the earlier material.

4 Expectations

- Please note, the use of cell phones is not permitted during lecture. These should be left in your backpack or not brought to class.
- Laptop computers may be used for taking notes or for in-class assignments but cannot be used for other activities during lecture.
- No student may record any classroom activity without the express written consent of the instructor. If you believe you need to record classroom activities as an accommodation, please contact the Office of Disability Resources (see Section ??).

Please keep in mind that these guidelines are necessary to maintain an environment that is safe and conducive for learning.

4.1 Expectations of Students and Instructors

The instructors and teaching assistants have the right to expect the following of students: (Adapted and modified from those developed by [Howard Culbertson at Southern Nazarene University](#)).

1. Students will arrive to class on time and will be prepared for the lecture.
2. Students will turn in assignments on time (see policy on assignment due dates).
3. Students will immediately inform the instructor or the teaching assistants if extenuating circumstances prevent the student from attending a lecture.
4. Students will follow the code of conduct regarding academic integrity, cheating, plagiarism, and collaboration as outlined in the syllabus.
5. Students will seek assistance when they need it.
6. If contacted by the instructor or teaching assistant, students will respond within 24 hours during the week and 48 hours on weekends.

The students have the right to expect the following of the instructor and teaching assistants:

1. A syllabus that describes class procedures, policies, and a course description will be provided.
2. Class sessions that will start and end on time.
3. Any changes to the course schedule will be provided to the students within 48 hours of the change.
4. The instructor will be available outside class either during their posted office hours or during other pre-arranged times.
5. If contacted by a student, the primary instructors or teaching assistants will respond within 24 hours during weekdays and 48 hours on weekends.

5 Academic Integrity

(See: <https://www.cmu.edu/policies/student-and-student-life/academic-integrity.html>)

Students in this course are expected to meet CMU's academic integrity standards in full — see the full university policy linked above. In short, this means submitting your own work, giving credit for others' ideas and contributions, and maintaining the trust that makes an academic community possible. Students who cannot meet these standards should voluntarily withdraw from this course.

5.1 Use of Generative AI in this course

To best support your own learning, you should complete all graded assignments in this course yourself, without any use of generative artificial intelligence (AI). Please refrain from using AI tools to generate any content (text, video, audio, images, code, etc.) for

an assignment or classroom exercise. Passing off any AI generated content as your own (e.g., cutting and pasting content into written assignments, or paraphrasing AI content) constitutes a violation of CMU's academic integrity policy³. If you have any questions about using generative AI in this course please email or talk to us.

5.2 Cheating and Plagiarism

In order to deter and detect plagiarism, online tools and other resources may be used in this class. It is the ethical responsibility of each student to identify the conceptual sources of any work submitted; failure to do so is dishonest and is the basis for a charge of cheating or plagiarism, which is subject to disciplinary action.

Cheating includes but is not necessarily limited to:

1. Plagiarism, explained below.
2. Submission of work that is not the student's own for papers, assignments, or exams.
3. Submission or use of falsified data.
4. Theft of or unauthorized access to an exam.
5. Use of an alternate, stand-in or proxy during an examination.
6. Use of unauthorized material including textbooks, notes or computer programs in the preparation of an assignment or during an examination.
7. Supplying or communicating in any way unauthorized information to another student for the preparation of an assignment or during an examination.
8. Collaboration in the preparation of an assignment. Unless specifically permitted or required by the instructor, collaboration will usually be viewed by the university as cheating. Each student, therefore, is responsible for understanding the policies of the department offering any course as they refer to the amount of help and collaboration permitted in preparation of assignments.
9. Submission of the same work for credit in two courses without obtaining the permission of the instructors beforehand.

Plagiarism includes, but is not limited to, failure to indicate the source with quotation marks or footnotes where appropriate if any of the following are reproduced in the work submitted by a student:

1. A phrase, written or musical.
2. A graphic element.
3. A proof.
4. Specific language.
5. An idea derived from the work, published or unpublished, of another person.

Any disciplinary actions regarding charges of cheating or plagiarism will follow the procedures of the home university of the student involved.

³<https://www.cmu.edu/policies/student-and-student-life/academic-integrity.html>

Collaboration vs. Cheating Collaboration is defined by Merriam-Webster's Collegiate Dictionary (10th edition) as "to work jointly with others or together, especially in an intellectual endeavor." You are encouraged to discuss the course material, concepts, and homework assignments with other students in the class. **However, each student must eventually submit their own unique work (e.g., homework solutions, final project).** If any collaboration was used to complete an assignment, record the names of the collaborators and the nature of the collaboration. Any attempt to submit work that is not the student's own work will be considered an act of cheating. In addition, any student who knowingly supplies their completed homework assignment to another student is violating the cheating policy and will be subject to disciplinary action.

ANY VIOLATION OF THIS POLICY WILL NOT BE TOLERATED AND THE PENALTY WILL BE FAILURE IN THE COURSE.

If you have any questions regarding this policy, contact the instructors.

6 Resources

6.1 Accommodations for Students with Disabilities

If you have an accommodations letter from the Office of Disability Resources, please discuss your accommodations and needs with me as early in the semester as possible so we can make sure they're in place. If you suspect you may have a disability and would benefit from accommodations but are not yet registered, contact the Office of Disability Resources directly: <https://www.cmu.edu/disability-resources/> (access@andrew.cmu.edu).

6.2 Statement of Support for Students' Health & Well-being

Take care of yourself. If you or anyone you know is experiencing academic stress, difficult life events, or feelings of anxiety or depression, please reach out for support sooner rather than later — to a friend, to me, or to Counseling and Psychological Services (CaPS): 412-268-2922, <http://www.cmu.edu/counseling/>.

If you or someone you know is feeling suicidal or in danger of self-harm, call someone immediately, day or night:

CaPS: 412-268-2922

Re:solve Crisis Network: 888-796-8226

Suicide and Crisis Lifeline: 988

If the situation is life threatening, call the police

On campus: CMU Police: 412-268-2323

Off campus: 911

6.3 Diversity Statement

We must treat every individual with respect. Diversity — of race, color, national origin, language, sex, disability, age, sexual orientation, gender identity, religion, creed, ancestry, belief, veteran status, genetic information, and more — is fundamental to building and maintaining an equitable and inclusive campus community, and each of us is responsible for creating a safer, more inclusive environment. See CMU's full diversity and inclusion statement and resources at <https://www.cmu.edu/inclusive-excellence/>.

If you experience or observe unfair or hostile treatment on the basis of identity, please speak out; the **Center for Student Diversity and Inclusion** (csdi@andrew.cmu.edu, (412) 268-2150) is here to help.

6.4 Ethics Reporting Hotline

To report suspected unethical or illegal activity related to academic and student life, financial matters, human relations, or campus safety — confidentially or anonymously — use CMU's Ethics Reporting Hotline, operated by a third party (NAVEX): call 1-844-587-0793 or visit <https://cmu.ethicspoint.com>. This hotline is *not* an emergency service; for emergencies, call CMU Police (412-268-2323) or 911.